

HUALIEN AP, Taiwan

WMO: 467630

Lat: 24.023N

Lon: 121.618E

Elev: 52

StdP: 14.67

Time Zone: 8.00 (E08)

Period: 86-99

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	53.1	55.0	46.0	46.0	56.3	48.0	49.7	57.3	25.4	61.9	23.1	62.0	5.3	240	0.320

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.4	91.5	82.1	90.0	81.5	89.5	81.3	83.8	89.1	82.9	88.3	82.2	87.7	8.9	150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
82.5	169.4	88.3	81.0	161.5	87.2	80.6	159.2	86.9	48.1	89.2	47.0	89.2	46.2	87.9	90.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
22.8	19.6	17.1	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
			46.0	86.1	2.8	2.0	44.0	87.5	42.3	88.7	40.7	89.8	38.6	91.3
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	74.1	64.4	64.4	67.8	72.9	77.1	80.7	83.3	83.3	80.2	76.2	71.3	66.7
	DBStd	7.78	4.17	4.81	5.54	4.84	3.62	3.03	1.94	2.00	2.81	2.95	3.90	4.12
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	200	61	64	36	4	0	0	0	0	0	0	5	30
	CDD50	8790	446	404	553	687	842	922	1031	1031	905	813	640	516
	CDD65	3515	42	47	124	241	377	472	566	566	455	348	195	82
	CDH74	30046	75	106	425	1324	2791	4846	6791	6643	4204	2048	629	164
	CDH80	9761	1	3	42	255	669	1658	2767	2684	1226	369	78	9
(14) Wind	WSAvg	6.1	6.8	6.4	6.1	5.7	5.4	5.3	5.2	5.4	5.9	7.0	7.4	7.0
(15) Precipitation	PrecAvg	83.8	2.6	4.0	3.5	4.4	8.0	9.0	7.0	9.3	13.9	14.1	5.2	2.1
	PrecMax	140.1	7.9	13.3	11.4	14.5	23.6	27.5	31.1	26.7	38.0	66.9	16.9	5.6
	PrecMin	38.9	0.5	0.9	0.9	1.0	1.2	1.0	0.6	0.1	1.3	0.5	0.2	0.6
	PrecStd	21.9	1.4	3.0	2.0	3.1	4.8	6.2	6.4	7.4	9.4	14.8	4.4	1.1
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	78.9	80.2	83.9	88.0	88.8	91.2	92.7	93.4	90.0	88.1	84.4	80.9
		MCWB	71.7	73.0	75.8	79.0	80.7	81.8	82.7	83.1	81.6	79.4	77.6	73.1
	2%	DB	75.8	77.1	80.8	84.7	87.1	89.7	91.5	91.5	89.2	85.8	82.4	77.5
		MCWB	69.3	70.6	74.2	77.0	79.6	81.2	82.1	82.1	81.2	78.3	75.6	70.6
	5%	DB	73.6	75.1	78.7	82.7	85.6	88.2	90.0	90.0	87.8	84.0	80.3	75.5
		MCWB	67.8	69.3	73.3	76.3	78.9	81.0	81.6	81.5	80.8	77.1	73.7	69.2
	10%	DB	71.4	72.0	76.6	80.7	84.0	87.5	89.5	89.4	86.2	82.1	77.3	73.5
		MCWB	66.4	67.4	71.7	75.0	77.9	80.8	81.4	81.2	79.9	76.0	72.2	67.8
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	72.3	73.8	77.3	79.8	82.4	84.1	84.9	85.0	83.9	81.1	78.9	74.3
		MCDB	77.3	78.4	82.1	85.6	87.0	88.3	89.8	90.2	88.4	85.3	83.0	79.2
	2%	WB	70.2	71.6	75.3	78.4	80.5	82.7	83.8	83.7	82.4	79.5	76.9	71.8
		MCDB	74.7	76.0	79.9	83.8	85.3	87.4	89.3	89.3	87.4	83.9	81.3	76.5
	5%	WB	68.7	69.8	73.5	77.0	79.3	81.8	82.8	82.7	81.0	78.2	75.0	70.0
		MCDB	72.8	74.0	77.8	82.3	84.0	86.8	88.6	88.6	86.1	82.4	78.7	74.3
	10%	WB	67.0	68.1	71.8	75.5	78.4	80.8	82.0	81.9	80.1	76.9	73.2	68.6
		MCDB	70.7	71.8	75.8	80.1	82.9	86.0	87.9	87.8	85.0	80.7	76.4	72.7
(35) Mean Daily Temperature Range	5% DB	MDBR	11.4	10.7	11.2	11.3	11.1	11.9	12.4	12.6	11.6	11.3	11.1	11.6
		MCDBR	15.4	15.2	14.4	13.9	13.0	13.0	13.4	13.8	13.3	13.9	14.3	15.0
		MCWBR	11.2	11.3	10.9	10.0	9.2	9.0	8.9	8.8	8.8	9.1	9.7	10.6
	5% WB	MCDBR	14.3	14.4	13.9	13.2	12.5	12.3	12.8	13.0	12.5	12.6	12.9	13.7
		MCWBR	11.1	11.1	11.1	9.9	9.3	9.0	9.3	9.1	8.9	9.0	9.7	10.7
(40) Clear-Sky Solar Irradiance	taub		0.541	0.576	0.697	0.703	0.613	0.551	0.528	0.561	0.559	0.543	0.532	0.547
	taud		1.899	1.857	1.623	1.662	1.886	2.054	2.124	2.013	2.004	1.998	2.001	1.914
	Ebn at Noon		224	227	207	210	229	242	248	239	237	233	226	216
	Edh at Noon		56	62	82	81	64	54	50	56	56	53	50	54
(44) All-Sky Solar Radiation	RadAvg		756	881	1002	1117	1308	1554	1813	1630	1429	1188	874	748
	RadStd		109	143	145	175	215	183	115	161	152	126	93	115

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (25 neighbors)	N/A	N/A	N/A	+0.49	N/A	N/A	N/A	N/A	N/A	+146

Nomenclature: See separate page